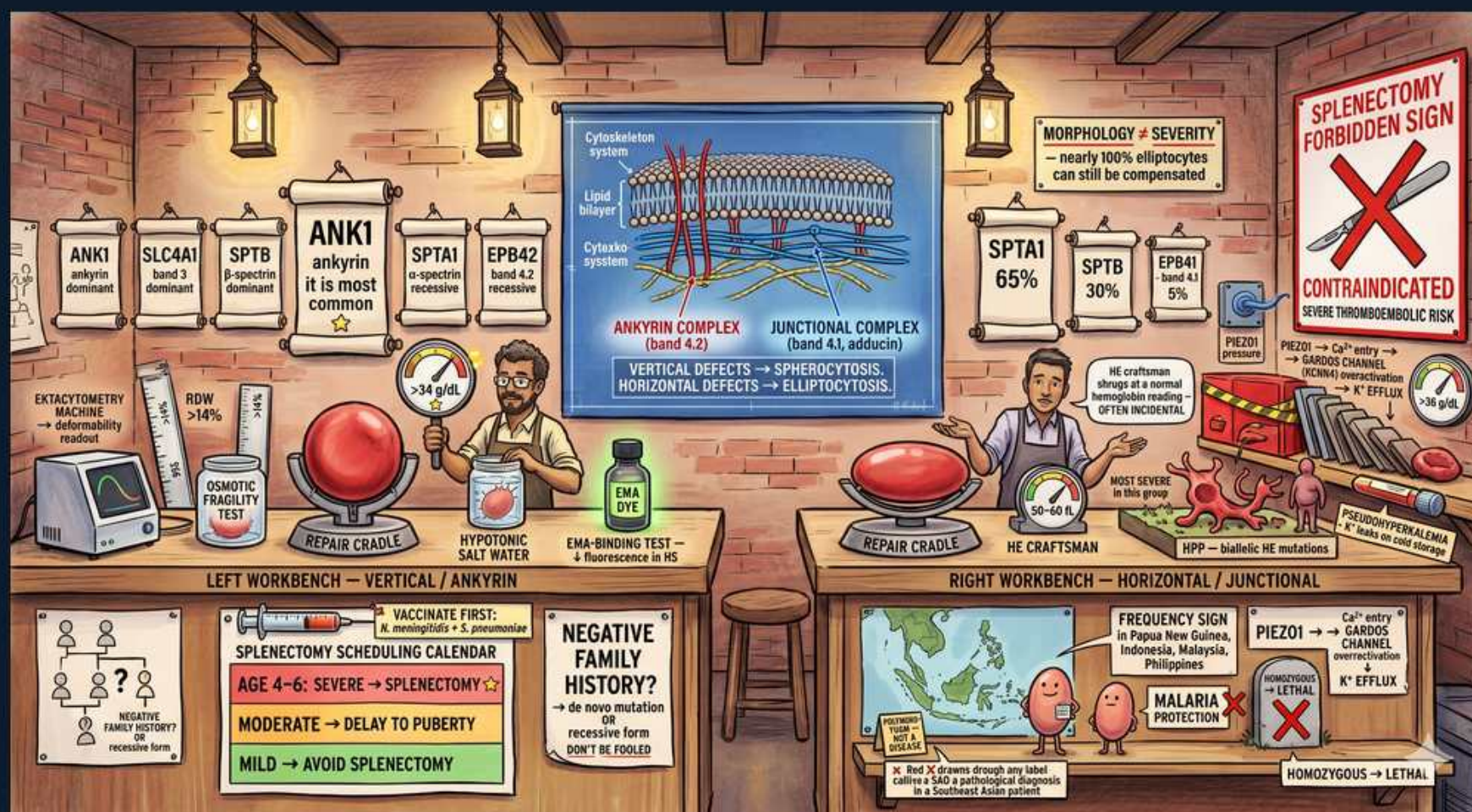


Hereditary Spherocytosis & Elliptocytosis — Membrane Defects



1. ANKYRIN (ANK1) — most common

WHAT YOU SEE

Star-marked ANK1 scroll, autosomal dominant

WHAT IT ENCODES

Ankyrin (ANK1) defect is the single most common cause of hereditary spherocytosis (~50%), usually autosomal dominant. Ankyrin links the spectrin cytoskeleton to band 3 in the lipid bilayer. Loss of vertical membrane connections sheds membrane, producing spherocytes.

BOARD TRAP

Don't pick spectrin as 'most common' — spectrin (SPTA1/SPTB) defects exist but ankyrin is the leading single gene; spectrin alone is more associated with elliptocytosis when horizontal.

2. BAND 3 (SLC4A1)

WHAT YOU SEE

SLC4A1 band 3, autosomal dominant

WHAT IT ENCODES

Band 3 (anion exchanger, SLC4A1) is the transmembrane anchor for ankyrin. Its deficiency reduces eosin-5-maleimide (EMA) dye binding — the basis of the EMA flow cytometry test for HS. Band 3 loss also underlies the 'pincerred' cells.

BOARD TRAP

EMA binds band 3 specifically; a low EMA result points to HS, not to enzymatic causes like G6PD or PK deficiency.

3. SPECTRIN (SPTA1/SPTB)

WHAT YOU SEE

SPTA1 / SPTB scrolls, recessive vs dominant

WHAT IT ENCODES

Alpha-spectrin (SPTA1, recessive) and beta-spectrin (SPTB, dominant) form the horizontal cytoskeletal lattice. Horizontal defects in spectrin self-association cause hereditary elliptocytosis; combined severe loss causes pyropoikilocytosis.

BOARD TRAP

Vertical defects (ankyrin/band 3) → spherocytes; horizontal defects (spectrin/protein 4.1) → elliptocytes. Mixing the two axes is a classic exam trap.

4. PROTEIN 4.2 (EPB42)

WHAT YOU SEE

EPB42 protein 4.2 scroll

WHAT IT ENCODES

Protein 4.2 (EPB42) stabilizes the ankyrin–band 3 junction. Its deficiency is a less common, often recessive HS — more prevalent in Japanese populations. Part of the vertical interaction complex.

BOARD TRAP

4.2 (EPB42) is vertical/HS; protein 4.1 (EPB41) is horizontal/elliptocytosis — the numbers are nearly identical so read carefully.

5. PROTEIN 4.1 → elliptocytosis

WHAT YOU SEE

Junctional complex band 4.1, adducin, HE craftsman

WHAT IT ENCODES

Protein 4.1 (band 4.1) anchors the junctional complex (spectrin–actin–adducin). Defects here cause hereditary elliptocytosis via horizontal lattice failure. Most HE is mild/compensated.

BOARD TRAP

HE from 4.1 is usually asymptomatic — don't over-treat. Severe hemolysis suggests hereditary pyropoikilocytosis (HPP), not simple HE.

6. Vertical defect → spherocytosis

WHAT YOU SEE

Blue label: vertical defects = spherocytosis

WHAT IT ENCODES

Vertical interactions tether the cytoskeleton to the bilayer (ankyrin–band 3–4.2). Their loss releases unsupported membrane, the spleen pits it, and the cell rounds into a spherocyte with increased osmotic fragility.

BOARD TRAP

Spherocytes also occur in warm autoimmune hemolytic anemia — distinguish with a DAT/Coombs (positive in AIHA, negative in HS).

7. Osmotic fragility test

WHAT YOU SEE

Blood-pressure-cuff 'machine', RDW >14%

WHAT IT ENCODES

Increased osmotic fragility confirms membrane loss: spherocytes lyse at higher saline concentrations than normal biconcave cells. Elevated MCHC and RDW support HS. Now largely supplanted by EMA flow cytometry.

BOARD TRAP

Osmotic fragility can be falsely normal in reticulocytosis or recent transfusion; EMA binding is more sensitive/specific.

8. EMA flow cytometry

WHAT YOU SEE

EMA-binding test, fluorescence in HS

WHAT IT ENCODES

Eosin-5-maleimide binds band 3 and related proteins; reduced fluorescence indicates HS (and CDA II). It is the preferred confirmatory test — sensitive and fast.

BOARD TRAP

Low EMA is not specific to a single gene — it is reduced across band 3, ankyrin, spectrin, and 4.2 defects.

9. Morphology vs severity

WHAT YOU SEE

Banner: ~100% elliptocytes can still be compensated

WHAT IT ENCODES

The percentage of abnormal cells on smear does not predict clinical severity. Nearly 100% elliptocytes in HE can be fully compensated with no anemia, whereas a few spherocytes may accompany brisk hemolysis.

BOARD TRAP

Don't grade disease by smear percentage; grade by hemolysis markers (hemoglobin, reticulocytes, bilirubin, LDH, haptoglobin).

10. Splenectomy — vaccinate first

WHAT YOU SEE

Calendar: vaccinate first, age timing

WHAT IT ENCODES

Splenectomy reduces hemolysis by removing the site of spherocyte destruction. Give pneumococcal, meningococcal, and Hib vaccines ≥ 2 weeks before surgery. Age 4–6 severe $\rightarrow$ splenectomy; moderate $\rightarrow$ delay to puberty; mild $\rightarrow$ avoid.

BOARD TRAP

Never splenectomize without prior vaccination — overwhelming post-splenectomy infection (encapsulated organisms) is the lethal risk.

11. Splenectomy timing by severity

WHAT YOU SEE

Age 4-6 severe; moderate delay; mild avoid

WHAT IT ENCODES

Mild HS: avoid splenectomy. Moderate: delay until at least puberty/age 6 to preserve immunity. Severe transfusion-dependent disease: splenectomy after age 4–6.

Cholecystectomy may be combined if pigment gallstones present.

BOARD TRAP

Pigment gallstones from chronic hemolysis are common — screen with ultrasound; gallstones in a young patient should prompt a hemolysis workup.

12. Splenectomy contraindicated risk

WHAT YOU SEE

Red 'SPLENECTOMY FORBIDDEN' sign, thrombosis

WHAT IT ENCODES

Post-splenectomy raises lifelong thromboembolic and infection risk; it is relatively contraindicated in mild disease and very young children. Persistent spherocytes and Howell-Jolly bodies appear on smear afterward.

BOARD TRAP

Howell-Jolly bodies after splenectomy are expected — don't mistake them for new pathology or non-functioning accessory spleen failure.

13. Negative family history

WHAT YOU SEE

Banner: 'don't be fooled' by negative family history

WHAT IT ENCODES

Up to ~25% of HS is from de novo or recessive mutations, so a negative family history does NOT exclude it. New spontaneous mutations and recessive inheritance explain sporadic cases.

BOARD TRAP

Absent family history is a distractor — it does not rule out hereditary spherocytosis.

14. Southeast Asian ovalocytosis (band 3)

WHAT YOU SEE

Frequency map: Papua New Guinea, Malaysia, Indonesia

WHAT IT ENCODES

A band 3 (SLC4A1) deletion variant causes Southeast Asian ovalocytosis, geographically clustered in Malaysia/Indonesia/PNG. Heterozygotes are mild and confer malaria protection; homozygosity is lethal.

BOARD TRAP

This band 3 variant is protective against malaria like HE/HS membrane variants — homozygous SAO is embryonic-lethal, so all patients are heterozygous.

15. Malaria protection / homozygous lethal

WHAT YOU SEE

Malaria mosquito + 'homozygous = lethal'

WHAT IT ENCODES

Membrane variants (elliptocytosis/ovalocytosis) impede Plasmodium invasion, giving heterozygote malaria protection — a balanced polymorphism in endemic regions.

Homozygous severe membrane loss is incompatible with life.

BOARD TRAP

Malaria protection is the heterozygote advantage; this is why these variants persist, paralleling sickle trait and G6PD.